

Suman Shrestha

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1 Education

Ph.D. in Physics 2021–Present

Temple University, Philadelphia, PA, USA

Advisor: Prof. Nikos Sparveris

Expected completion: 2027–2028

M.Sc. in Physics 2017–2021

Tribhuvan University, Kathmandu, Nepal

B.Sc. in Physics 2012–2016

Tribhuvan University, Kathmandu, Nepal

2 Current Research

My current research focuses on analyzing CLAS12 RG-D semi-inclusive deep-inelastic scattering (SIDIS) data at Jefferson Lab to investigate nuclear transverse-momentum-dependent distributions (nTMDs), quark propagation, and hadronization in nuclei. I study hadron multiplicity ratios, transverse-momentum broadening, and azimuthal modulations across different nuclear targets.

3 Research Experience

Graduate Research Assistant August 2023–Present

Temple University, Philadelphia, PA

- Conducted a research rotation in Summer 2023, developing a theoretical model for Mott scattering to study electron-nucleus interactions, performing data analysis and plotting using Mathematica, and presenting findings at the Temple University Graduate Symposium in August 2023.
- Analyzed experimental data from Jefferson Lab for nuclear Transverse Momentum Distributions (nTMDs), focusing on quark dynamics in nuclei with targets like liquid deuterium (LD2), carbon (CxC), and copper-tin (CuSn), and developed a particle identification model to enhance data accuracy.
- Played a key role in a team calibrating data for the Central Vertex Tracker across multiple experiments and developing Drift Chamber calibration software for the ALERT detector at Jefferson Lab, supporting ongoing and future nuclear physics research.
- Collaborated with graduate students, postdocs, professors, and scientists from various institutions and Jefferson Lab, presenting research updates at Jefferson Lab collaboration meetings and contributing to experiments through data collection and service work.
- Participated in the MUSE experiment at the Paul Scherrer Institute (PSI), Switzerland, in July 2024 and October 2025, contributing to data collection for precision measurements of the proton charge radius.

4 Publications

- [1] A. Accardi, P. Achenbach, D. Adhikari, A. Afanasev, C.S. Akondi, N. Akopov, et al., “Strong interaction physics at the luminosity frontier with 22 GeV electrons at Jefferson Lab,” *Eur. Phys. J. A* **60**(9), 173 (2024).

- [2] S. Adhikari, B.A. Raue, D.S. Carman, L. Guo, T. Chetry, P. Achenbach, et al., “Measurement of beam-recoil observables C_x and C_z for $K^+\Lambda$ photoproduction,” *Phys. Rev. C* **112**(6), 065211 (2025).
- [3] A. Kripko, S. Diehl, K. Joo, P. Achenbach, J.S. Alvarado, M. Amarian, et al., “Multidimensional measurements of beam single-spin asymmetries in semi-inclusive deep-inelastic charged-kaon electroproduction off protons in the valence region,” *Phys. Rev. C* **112**(5), 055202 (2025).
- [4] M. McEneaney and A. Vossen et al., “Longitudinal Spin Transfer to Λ Hyperons in Semi-Inclusive Deep Inelastic Scattering with CLAS12,” *arXiv preprint* arXiv:2509.05758 (2025).
- [5] D.S. Carman, A. d’Angelo, L. Lanza, V.I. Mokeev, P. Achenbach, J.S. Alvarado, et al., “Recoil polarization in K^+Y electroproduction in the nucleon resonance region with CLAS12,” *Phys. Rev. C* **112**(3), 035206 (2025).
- [6] P. Roy, S. Cao, V. Crede, E. Klempt, V.A. Nikonov, A.V. Sarantsev, V.D. Burkert, et al., “Measurement of single- and double-polarization observables in the photoproduction of $\pi^+\pi^-$ meson pairs off the proton using CLAS at Jefferson Laboratory,” *Phys. Rev. C* **112**(3), 035201 (2025).
- [7] H. Jiang, S. Strauch, I.I. Strakovsky, D. Rönchen, R. Workman, et al., “Polarization Observables T and F in the $\gamma p \rightarrow \pi^0 p$ Reaction at CLAS,” *arXiv preprint* arXiv:2508.17589 (2025).
- [8] S. Dytman, M. Betancourt, N. Steinberg, L.B. Weinstein, A. Ashkenazi, et al., “Proton Transparency and Neutrino Physics: New Methods and Modeling,” *arXiv preprint* arXiv:2508.01905 (2025).
- [9] V. Klimenko, D.S. Carman, R.W. Gothe, K. Joo, N. Markov, V.I. Mokeev, et al., “Inclusive electron scattering in the resonance region off a hydrogen target with CLAS12,” *Phys. Rev. C* **112**(2), 025201 (2025).
- [10] A.V. Sarantsev, E. Klempt, K.V. Nikonov, P. Achenbach, V.D. Burkert, V. Crede, et al., “Photoproduction of two charged pions off protons in the resonance region,” *Phys. Rev. C* **111**, 035203 (2025).
- [11] H.T. Klest, S. Joosten, H. Szumila-Vance, W. Armstrong, F.A. Flor, B. Kim, et al., “Studying the Strangeness D -Term in Hall C via Exclusive ϕ Electroproduction,” *arXiv preprint* arXiv:2501.01582 (2025).
- [12] P. Chatagnon, V. Kubarovsky, R. Paremuzyan, S. Stepanyan, M. Tenorio, et al., “Measurement of the near-threshold J/ψ photoproduction cross section with the CLAS12 experiment,” *arXiv:2602.22128 [nucl-ex]* (2026).
- [13] A.G. Acar, P. Achenbach, J.S. Alvarado, M. Amarian, W.R. Armstrong, H. Atac, et al., “Measurements of Beam Spin Asymmetries of $\pi^\pm\pi^0$ dihadrons at CLAS12,” *arXiv preprint* arXiv:2602.14271 (2026).
- [14] R. Ratvasky, T. Rostomyan, M. Ali, H. Atac, F. Barchetti, J.C. Bernauer, et al., “The MUSE Target Chamber Post Veto,” *arXiv preprint* arXiv:2602.10279 (2026).
- [15] S.J. Paul, M. Arratia, H. Hakobyan, W. Brooks, A. Acar, P. Achenbach, et al., “First Study of the Nuclear Response to Fast Hadrons via Angular Correlations between Pions and Slow Protons in Electron–Nucleus Scattering,” *Phys. Lett. B*, 140250 (2026).
- [16] H. Atac, A. Camsonne, M.K. Jones, M. Paolone, N. Sparveris, N. Sayadat, et al., “Measurement of the generalized polarizabilities of the proton in virtual Compton scattering,” *arXiv preprint* arXiv:2308.07197 (2023).
- [17] W.R. Armstrong, S. Lee, B. Kim, M. Kim, H.T. Klest, V. Klimenko, Z.E. Meziani, et al., “A Measurement of the Coherent J/ψ Electroproduction Cross Section off ^4He (PAC53),” Jefferson Lab PAC 53 Proposal (2025).

- [18] A. Arora, H. Chinchay, M. Farooq, C. Lama, E. Long, M. McClellan, et al., “Revealing the Transition Region of QCD with the Proton’s g_2 Structure Function (PAC53),” Jefferson Lab PAC 53 Conditional Proposal, submitted May 4, 2025.

5 Talks

- [1] “Study of Nuclear TMDs in SIDIS with CLAS12 at Jefferson Lab,” APS Global Physics Summit, March 17, 2026.
- [2] “Study of Nuclear TMDs in SIDIS with CLAS12 at Jefferson Lab,” CLAS Collaboration Meeting, March 2026.
- [3] “Study of purity in the pion identification with and without LTCC,” CLAS Collaboration Meeting, Online Only, November 19, 2025.
- [4] “Study of nuclear TMDs in SIDIS with CLAS12 at Jefferson Lab,” Temple Physics Research Symposium, Temple University, September 15, 2025.
- [5] “RG-D: PID and Contamination Studies,” CLAS Collaboration Meeting, Christopher Newport University, Newport News, VA, July 10, 2025.
- [6] “First Look at Charged Pions SIDIS Kinematics on RG-D Data,” CLAS Collaboration Meeting, Jefferson Lab, Newport News, VA, November 15, 2024.
- [7] “First Look at MVT High Voltage Tables and RG-D Data,” CLAS Collaboration Meeting, Christopher Newport University, Newport News, VA, June 25–28, 2024.

6 Posters

- [1] “Study of Nuclear TMDs for Different Nuclear Targets at CLAS12,” Temple University Research Symposium, August 26, 2024.

7 Teaching Experience

Graduate Teaching Assistant

Aug. 2021–May 2023; Jan. 2025–Apr. 2025

Temple University, Philadelphia, PA

- Mentored over 500 undergraduate students in physics laboratory courses, covering mechanics, electricity, magnetism, optics, and thermodynamics.
- Supervised experimental setups, guiding students in data collection and analysis using Capstone and Excel.
- Collaborated with other Teaching Assistants, the lab coordinator, and professors to revise and update instructional materials, including lab manuals and grading rubrics.
- Explained theoretical concepts and their connection to real experiments, helping students understand physics principles.
- Proctored exams for large enrollment courses, ensuring academic integrity.

8 Awards and Honors

Stanislav Kotsev Memorial Award

2025–2026

Department of Physics, Temple University

Awarded to a graduate student of exceptional character making excellent progress in the Physics Graduate Program.

University Grants Commission (UGC) Award

2019

Tribhuvan University, Nepal

Recipient of a government-funded grant supporting graduate students’ research, awarded for my master’s thesis in physics.

9 Memberships

- CLAS Collaboration, Jefferson Lab, Newport News, VA, USA
- MUSE Collaboration, Paul Scherrer Institute (PSI), Villigen, Switzerland

10 Computing Skills

- **Languages:** Python, C++, Java, Bash
- **Scientific Tools:** ROOT, Geant4, clas12root, LaTeX
- **Version Control / Systems:** Git, Linux, macOS, Slurm